

COC EXAM FOR BSC NURSE

1. A 50-year-old women patient diagnosed with early rheumatoid arthritis of the left hand and to see the progress of the disease, her physician sent her to the radiology department for hand x-ray.

Which is the best positioning to see the progress?

- A. lateral
- B. Postero-Anterior of the left hand
- C. Postero-Anterior of the Right hand
- D. Postero -Anterior of both hands

2. A patient has an examination of chest x-ray in x-ray department with the clinical history of shoulder dislocation but Technolgist cannot make a patient in basic Anterio - posterior Supine or Erect position b/c the patient can't even slightly Abducted his Arm.

what is the best alternative position?

- A. Superio - inferior
- B. Infero - Superior
- C. PA Oblique (Y) Projection
- D. Antero-posterior oblique humerus

3. A 50-year-old patient come to the x-ray department for Abdominal x-ray with history of Nausia, vomiting, Abdominal distension and pain. The Abdominal x-ray showed that dilated bowl loop with air fluid level.

What is the possible diagnosis for this patient?

- A.Hital Hernia
- B.Bowl obstruction
- C.Abdomenal mass
- D.Acute Appendicitis

Based on question #3, which is the best positioning to rule out your diagnosis?

- A. PA-prone
- B. AP-Erect
- C. AP-supine
- D. Posterior oblique

4. Based on question #3, which is the best positioning to rule out your diagnosis?

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5. A patient has an examination of chest x-ray; the clinical history of the patient is left side pneumothorax (air in the pleural space) but you cannot make a patient in basic position.

what is the best alternative position?

- A. A patient in erect position
- B. Patient in supine position
- C. A patient in Left lateral and cassette at the back
- D. A patient in Right lateral and cassette at the back

6. A 43-year-old male patient came to x-ray room with a history of chest pain, severe dyspnea, tachycardia and hypotension and Distended neck veins. Chest x-ray shoed that increased lucency, ipsilateral increased intercostal spaces, contralateral shift of the mediastinum and depression of the hemidiaphragm.

What is the most diagnosis for this case?

- a. Severe lung collapse

- b. Tension pneumothorax
 - c. Massive pleural effusion
 - d. Huge parenchymal bullae
7. A patient came to the x-ray department for a chest x-ray with clinical indication of suspected rib fracture which one of the following would not be appropriate for this specific patient?
- A) A pa erect chest x-ray with deep inspiration and oblique projection should be done
 - B) A pa and oblique projections without forced inspiration should be taken
 - C) An AP and lateral chest x-ray would be sufficient
 - D) A lateral chest x-ray would be sufficient
8. A 53 years old patient came to the x-ray department for an IVU procedure which of the following would be correct sequence of images to be taken?
- A) Nephrogram --- prone image --- supine film with compression ---- supine film (release)
 - B) Nephrogram --- Supine film with compression ---supine film (release) --- prone image
 - C) Supine film with compression --- Nephrogram --- prone image --- supine film (release)
 - D) Nephrogram --- supine film with compression --- prone image ---- supine film (release)
9. A 50-year-old women patient diagnosed with early rheumatoid arthritis of the left hand and to see the progress of the disease, her physician sent her to the radiology department for hand x-ray.

Which is the best positioning to see the progress?

- A.lateral
- B.Postero-Anterior of the left hand
- C.Postero-Anterior of the Right hand

D. Postero -Anterior of both hands

10. A patient came to the x-ray department for a skull x-ray based on the ALARA principle what technical measure would markedly increase the patient dose?

A) increasing the milli ampere second (MAS)

B) decreasing the exposure time

C) increasing the KVP

D) decreasing the KVP

11. A patient came to the radiology department for a chest x-ray and a Postero Anterior (PA) projection was done on image evaluation what technical factors have to be considered?

A) The Scapula should be seen projected over the upper poles

B) On PA chest x-ray magnification of the cardiac shadow is expected.

C) A diffusely over penetrated image would be expected to appear more opaque.

D) A visualization of 8-10 posterior ribs above the diaphragm would suggest a good inspiration.

12. A young patient comes to the x-ray department for calcaneus (axial) view all of the following techniques should be done for appropriate visualization except.

A) patient is supine or seated with the affected limb extended.

B) The central ray is angled 40° cephalad from the long axis of the foot centred at the base of the 3rd metatarsal (midfoot)

C) The patient should lie laterally on the affected foot side and lateral side of the affected foot should rest on the image receptor.

D) Foot is dorsiflexed until the plantar surface is running perpendicular to the image receptor this can be aided with tape or fabric wrapped around the distal phalanges to be pulled backwards by the patient

13. A 65 years old patient with a history of prolonged use of Non-steroidal anti-inflammatory drugs (NSAIDs) and recent frequent onsets of peptic ulcer disease came to the radiology department for an abdominal x-ray which of the following procedures is appropriate for this patient?

- A) Alateral abdominal x-ray should be done
- B) A supine abdominal x-ray would be sufficient
- C) A supine x-ray with lifted legs should be performed.
- D) An erect abdominal x-ray or a decubitus x-ray with AP projection of the beam

14) A highly fatigued 70-year-old patient with anosmia and headache came to the x-ray department for a chest x-ray. His radiograph showed bilateral airspace opacities (consolidation) with peripheral and lower zone predominance. No evidence of pleural effusion was noted. Which of the following abnormalities would fit the description above

- A) Emphysema
- B) SARS-COV-2
- C) Pneumothorax
- D) Bilateral lower lobe collapse

15) A patient came to the x-ray department with an indication of acute sinusitis and waters view was requested what technical factors should be implemented?

- A) 24 cm x 30 cm cassette should be used.
- B) the patient is erect facing the upright detector

C) the chin is raised until the menton-mandibular line (MML) is perpendicular to the receptor (OML will be 37° from receptor)

D) The occipital bone of the patient should come in contact with the image receptor with the x-ray beam projected anterior posteriorly.

16) on the radiograph acquired to the patient mentioned in question number 8 the radiograph acquired should have the following characteristics except

A) The coronoid process should be symmetrical.

B) The petrous ridge should be inferior to the maxillary sinuses

C) The base of the mandible should be superimposed with the occiput.

D) The petrous ridge should be exactly superior to the maxillary sinuses.

17) A 6-year-old patient came to the x-ray department after a road traffic accident his femoral radiograph showed a crack only on the lateral side of the femur with normal looking skin. which of the following types of fracture best describes it?

A) comminuted fracture

B) Open compound fracture

C) simple fracture

D) Greenstick fracture

18. A 26 years old female patient comes in with fertility problems in your department. The physician orders a hysterosalpingogram. When is the best time to perform this procedure?

A. Right before her period

B. Two days after her period has started

C. Three days after her period has ended

D. Any time during period

19. A 35 years old male patient comes with clinical history of renal failure with total anuria, ultrasound was found to be normal, what is your next consider regarding investigation?

- A. Doppler Ultrasound
- B. KUB
- C. IVP
- D. RFT

20. A 42 years old male patients comes with history of road traffic accident to your radiology department suspected with long bone (thigh) fracture & the physician requests you to do x ray, what a crucial measure you are going to take to perform this procedure among the following?

- A. Do PA & Lateral x ray
- B. Position the Pt properly
- C. Consider anode heel effect
- D. Select appropriate 35 * 43 film size

21. A 25 years old Female patient comes with history of falling down accident & the Rt side of the chest is hyper-resonant to percussion, then the physician order PA chest x ray of this patient, so what x ray finding would you expect to be present on the film?

- A. Air filled bronchi running through fluid – filled alveoli

B. An increase in Broncho vascular marking & bronchi seen end which may appear as ring shadows

C. Flattened hemi diaphragm with increased & irregular radiolucency of the lung

D. Radiolucent lung with absent vascular markings & ill-defined lung edge

22. A 2 days old neonate with excessive salivation develops respiratory distress. Attempts to pass an or gastric fail because the catheter coils in the back of the throat. A chest film is obtained & shows right upper lobe atelectasis and a gasless abdomen. The most likely diagnosis is:

A. Proximal esophageal atresia without a fistula

B. Proximal esophageal atresia with a distal tracheoesophageal Fistula

C. "H" Type TE fistula

D. Congenital Esophageal stricture

E. Esophageal atresia with both proximal & distal TE fistula

23. A 36 years old female patient presents with impaired vision, diplopia, severe Headaches and other neurologic symptoms. After her clinical examination, her physician suspect sellar mass (Pituitary adenomas) and she was sent to the Radiology department for skull radiography as initial investigation modality.

Which one of the following radiographic positioning of the skull is the best radiographic projection and beam angulation respectively?

A. Lateral, 20-degree caudal

B. Submentovertical, perpendicular to the film plate

C. Occipito-mental, perpendicular to the cassette holder

D. Lateral, perpendicular to the film plate

24. Patient on the supine position... the midsagittal plane of the patient is centered and perpendicular to the long axis of the table ...Depressed the patient chin and orbito - meatal line

is perpendicular to the film plane ... the central ray enters at a point approximately 5 cm above the glabella and exits through the Foramen magnum with 30° caudal beam angulation ...”

Which one of the following radiographic positioning best describes the above radiographic scenario?

- A. Submentovertical (Full basal skull)
- B. Stenver's view skull for Petrous Portion
- C. Mastoid- Lateral Oblique
- D. AP Axial (Towne) Skull

25. A 42-year-old female patient undergoes Excretory Urography with the complaint of dysuria, frequency, urgency, suprapubic pain and/or hematuria, but during the injection process some of the contrast medium seeps out of the vein to the surrounding tissue and causes localized edema.”

Which one of the following statements best describes the situation?

- A. Hypervolemia
- C. Hematuria
- B. Extravasation
- D. A vasovagal reaction

26. Just like in the above clinical scenario, if you observe the signs and symptoms of reactions to contrast media, what do you do next? What is your role as technologist?

- A. Continue the injection and call emergency team.
- B. Monitor vital signs and Administer cardiopulmonary resuscitation.
- C. Place the patient in Semi-fowler's position.
- D. Nothing is expected from the technologist

27. A 24-year-old woman patient comes to a hospital with the complain of epigastric (substernal pain) postprandial fullness, substernal fullness, nausea, and retching. Her contrast study of the stomach shows; part of the stomach protrudes through the esophageal hiatus of the diaphragm. What is the most likely diagnosis?

- A. Sliding hiatal hernia.
- B. Esophageal varices.
- C. Barrett's esophagus.
- D. Carcinoma of the Esophagus

28. In the above clinical scenario, what was the best patient positioning and radiographic projection respectively to better demonstrate the pathology?

- A. Erect & AP.
- B. Prone & PA.
- C. Erect & PA.
- D. Supine & AP.

29. A 26-year-old female patient is brought to the outpatient department with an acute renal colic, severe pain originating in the flank and radiating inferiorly and anteriorly. Her IVU shows distention of the calyces, renal pelvis and proximal ureter of unilateral kidney.

What is the most likely diagnosis?

- A. Hydronephrosis
- B. Ureterocele
- C. Horseshoe and Pelvic Kidney
- D. Renal cell Carcinoma

30. A 12 days' male neonate is presented with ultrasound description of idiopathic thickening of gastric pyloric muscle, which results in a progressive gastric outlet obstruction in infants with a clinical symptom of non-bilious projectile vomiting and palpable RUQ mass. What is the most likely finding?

- A. Gastritis
- C. Peptic ulceration
- B. Duodenal atresia
- D. HPS

31. A Physician has requested double contrast study for Upper GI Series. A Technologist has appropriately positioned the patient and is ready to start the scan.

What is the most correct sequence of events when performing the procedure?

- A. Patient is given gas-producing substance, then given a small amount of high density barium, then placed recumbent.
- B. Patient is placed recumbent, given a small amount of high-density barium, then given a gas-producing substance.
- C. Patient is given a gas-producing substance, placed recumbent, then given a small amount of high-density barium.
- D. Patient is given a small amount of high-density barium, placed recumbent, then given a gas-producing substance.

32. The patient was sent to the Radiology for the contrast study of the esophagus and during RAO of Esophageal, how much should the patient be inclined?

- A. 350- 400 from prone position
- B. 400 -700 from prone position
- C. 350 - 400 from supine position
- D. 400 -700 from supine position

33. A CXR of adult patient showed, air lucency under the right hemi diaphragm outlining the inferior surface of the right diaphragm and the dome of the liver. The patient has no direct or rebound abdominal tenderness. What is the possible differential diagnosis?

- A) Perforated peptic ulcer
- B) Inflammatory bowel disease
- C) Normal air
- D) Chalciditis syndrome

34. For congenital anomalies and valvular heart disease, the simplest and appropriate imaging of choice is

- A) Chest radiography
- B) echocardiography
- C) MRI
- D) CT

35. True about plain CXR

- A) PA view must be done whatever the patient condition is
- B) AP view shall be done for the children and critical patient
- C) Both AP and PA shall be combined for reduction of magnification
- D) No difference between AP and PA on x-ray interpretation

36. A new born infant sent for an x-ray from emergency department and the radiograph shows classic double bubble sign with gas filled distended stomach and duodenum with an absence of distal gas. What is the sign seen on the x-ray?

- A) Duodenal atresia
- B) Toxic megacolon

C) Anal atresia

D) Hirschsprung's disease

37. 25 years' female patient is sent to x-ray department following a fall. She felt sharp and stabbing pain in her right chest immediately after the fall. She also feels uncomfortable when inhaling air. Her physician requested Chest x-ray to rule out rib fracture.

What is the appropriate positioning used for this patient?

- A. AP chest x-ray
- B. PA chest x-ray
- C. Right anterior oblique
- D. Left anterior oblique

What is the appropriate film size used for this examination?

- A. 18 x 24 cm
- B. 24 x 30 cm
- C. 30 x 30 cm
- D. 35 x 35 cm

38. Mr. x is a radiographer who works at x-ray department which uses conventional x-ray machine. Mr. x is now in the dark room to process an exposed x-ray film.

Which one of the following is the appropriate step the Radiographer should follow to properly process the x-ray film?

- A. Developing → fixing → rinsing → washing → drying
- B. Developing → rinsing → fixing → washing → drying
- C. Developing → washing → rinsing → fixing → drying
- D. Developing → rinsing → washing → fixing → drying

In which film processing step does the latent image convert into a visible manifest image?

- A. Developing
- B. Drying
- C. Fixing
- D. Rinsing

39. 8 years old male patient is sent to x-ray department following accidental aspiration of a foreign body.

The aspirated coin is most likely to lodge in

- A. Esophagus
- B. Left main bronchus
- C. Right main bronchus
- D. Small intestine

40. Which of the following describes the relationship between esophagus and trachea?

- A. Esophagus is lateral to trachea
- B. Esophagus is posterior to trachea
- C. Trachea is posterior to esophagus
- D. Trachea is superior to esophagus

41. 45 years old patient came to the x-ray department with history of cough, breathlessness and reduced tolerance to exercise. The chest x-ray revealed bilaterally blunted costo-phrenic angles. Otherwise there are no other abnormal findings seen.

What is the most likely diagnosis?

- A. Ascites
- B. Bronchitis
- C. Emphysema
- D. Pleural effusion

42. 70 years old male patient with history of breathlessness and fatigue was sent to x-ray department. The patient was on wheelchair and unable to stand up. This patient had previous supine chest x-ray which revealed no abnormality. But the physician suspected pleural effusion.

Which one of the following is the sensitive projection used to detect minimum amount of pleural effusion?

- A. AP chest x-ray
- B. Prone chest x-ray
- C. Lateral decubitus
- D. Apical view

43. An accurately positioned oblique projection of the lumbar vertebral x-ray will demonstrate the “scotty dog” appearance, what bony structure does the scotty dog's nose represent?

- A. Pedicle
- B. Transverse process
- C. Pars interarticularis
- D. Superior articular process

44. A 40 year old woman patient presents with acute pain in the right upper quadrant for 3 hours duration. The pain radiates to her shoulder with associated nausea. The technologist was visualized a thickened gallbladder wall. The gallbladder wall shows alternate hypo and hyperechoic layers. The patient was very tender while scanning on Transverse scan of Transabdominal ultrasound.

What is the most likely diagnosis of this patient?

- A. Gb sludge
- B. cholelithiasis
- C. Adenomyomatosis
- D. Acute cholecystitis

45. while the technologist scanning a 30-year-old man with elevated bilirubin and lipase and a history of diarrhea. She incidentally found small echogenic well-defined mass in the right lobe of the liver. No vascularity was demonstrated in the lesion.

What is the most possible diagnosis for this patient?

- A. Pyogenic Abscess
- B. Liver hemangioma
- C. Hydrated cyst of Liver
- D. Focal nodular Hyperplasia

46. An 82-year-old-man presented with urinary retention and hematuria. The PSA blood level was 8 ng/ml. The technologies identified an enlarged prostate and the hypertrophied median lobe of prostate projects into the bladder lumen.

What will the technologist put as a diagnosis for this patient?

- A. Ureterocele
- B. Bladder calculus.

C. Bladder diverticulum

D. Benign prostatic hyperplasia

47. A pregnant mother is come to the ultrasound unit for her ANC follow up. While the technologist is scanning the pregnant mother, she suddenly got Nausea, sweating and pallor. What measurement should have the technologist take?

A. Avoid the situation and keep the scanning

C. scold her for ruined the Examination.

B. stop the scanning and Make another appointment

D. Wait until the patient feel better, then After start the scanning by placing the patient on her right/left lateral position

48. A 28-year-old male patient presents to the ultrasound department. He has a history of a sudden onset of abdominal pain, and an elevation in amylase and lipase. Sonographic findings include a hypoechoic region in the head of the pancreas and a small fluid collection adjacent to the pancreatic body. What is the most likely diagnosis?

A. Chronic pancreatitis

B. Focal acute pancreatitis

C. Pancreatic adenocarcinoma

D. Pancreatic cystadenocarcinoma

49. The sonographer was scanning 29-year-old Female patient with a history of menorrhagia & abdominal discomfort on Transabdominal Ultrasound Then she noticed a uterus demonstrating a myoma with a mass-like appearance and the endometrial echo interface was non visualized.

Which type of Myoma was the sonographer visualized?

A. Cervical

B. Interstitial

C. Submucosal

D. Pedunculated

50. A patient who had a previous history of renal stone came to ultrasound unit straight from emergency clinic for an abdominal ultrasound examination. On ultrasound evaluation, the technologist has noticed the left kidney to have a dilated renal pelvi calyceal system with anechoic urine.

What is the most likely diagnosis of this patient?

- A. Pyonephrosis
- B. Hydronephrosis
- C. Nephrocalcinosis
- D. Hemo-hydronephrosis

51. A pregnant woman is sent to ultrasound unit for obstetric ultrasound. During scanning of the uterus, the radiology technologist identified a tiny cystic structure implanted within the echogenic decidua at 5Wks of gestation using transabdominal scan (TAS).

What is the most likely structure seen in this case?

- A- Intradecidual sign
- B- Yolk sac
- C- Fetal pole
- D- Double sac sign

52. Mr. Obang present to emergency department of Gambella general hospital and the physician on duty sent for laboratory and sonographic examination. The lab result came with deranged liver function test and the sonographer impression is a heterogeneous rounded mass with solid fluid level internal echo debris on the right lobe of liver. On his presentation Mr. Obang has vomiting and mild fever. What's is the possible diagnosis?

- A) Amoebic abscess
- B) Hematoma
- C) Echinococcal cyst

D) ALL

53. A patient come with a history of chronic nocturnal cough and dyspnea. He feels better when he use multiple pillow at night and following his history a physician sent for abdominal ultrasound and there's ascites, right pleural effusion and prominently dilated hepatic Vein and IVC.what is the possible cause?

A) Primary biliary cholangitis

B) congestive cardiac failure

C) Budd Chiari syndrome

D) A and C

54. A 40 years old female patient come with a history of right upper quadrant pain and tenderness. She sent for sonographic study. the sonographer's revealed edematous gall bladder with wall halo sign and echogenic foci within the lumen causing strong acoustic posterior shadowing. The patient has fever and increased WBC count on laboratory finding. What is your possible diagnosis?

A) Acute Hepatitis

B) Acute pancreatitis

C) Acute cholecystitis

D) Acute pyelonephritis

55. A 28 Years old female come to the ultrasound unit with non-specific abdominal pain. There's mild ascites, hepatosplenomegaly as well as occlusion of the hepatic vein and thrombosis of inferior vena cava seen. What is the possible cause of occlusion and thrombus?

A) Budd Chiari syndrome

B) Cholangiocarcinoma

C) Hepatoma

D) Cholangitis

56 A 36 weeks' gravid mother come with painless light vaginal bleeding and on sonography there's partial coverage of internal cervical os by placental edges. The fetus has reassuring breath and it's on the good condition ...what is the possible cause of bleeding?

A) Placenta accreta

B))placenta Previa

C))abruption placenta

D) Vasa Previa

57. A 4-month neonate present to the emergency department with a projectile vomiting and the pediatrician sent for sonographic study. On sonography there's distended gastric lumen and thickened luminal wall were seen with increased pyloric canal length and antral nipple sign.as a sonographer what is your possible differential diagnosis?

A) Hypertrophic pyloric stenosis

B) Gastric volvulus

C) Necrotizing enter colitis

D) Duodenal atresia

58. For a routine antenatal OBGYN follow up a 32 weeks multigravida mother sent to a sonography department and as an individual working in the unit you saw on the sonography distended fetal bladder and distal ureter having a classic key hole sign with mild hydronephrosis.what is your possible impression?

A) Posterior urethral valves

B) PUJ obstruction

C) Unilateral renal agenesis

D) Ureterocele

59. A 30 years old female patient present to OBGYN clinic with complaints of chronic pelvic pain, dyspareunia, dysmenorrhea and menometrorrhagia. she sent for pelvic ultrasound study and sonography reveals bulky uterus with sub endometrial cysts and asymmetrical myometrium thickening and irregular myometrium-endometrial junction. What is the pathology seen on the uterus?

A) Adenomyosis

D) Endometrioma

C) Endometrial polyps

D) Myoma

60. An abdominal CT was requested for a 30 yr. old male patient who come to the radiology department with a complain of RUQ pain. The precontrast scan was done and you see a liver lesion as a technologist what would be your next move?

A. Let the patient go

B. Let the physician decide your next move

C. Give patient contrast n scan post contrast

D. Check on patient creatinine and scan post contrast

61. A 26 wks pregnant woman sent to CT exam with the history of server trauma to the head.

The physician wants to r/o skull fracture and internal hemorrhage. As a technologist what would you do to decrease the patient radiation dose besides shielding the patient.

A. Increase pitch

B. Decrease pitch

C. Increase slice thickness

D. Decrease slice thickness

62. A sinus CT was requested for a patient with a dental filling. Which one of the following two strategies can reduce the effect of the dental filling in the acquired image?

- A. high mA and short scan time
- B. small focal spot and small display field
- C. low kVp and high contrast or bone algorithm
- D. adjust the angle of the gantry and widen the width

63. CT scan was done for a 84 yrs old patient and in the CT pinecone shaped calcified structure was seen, what would be the most likely brain structure seen

- A. Pineal gland
- B. Choroid plexus
- C. Caudate nucleus
- D. Middle cerebral artery

64. A physician has requested a routine brain CT examination to be performed. A technologist has appropriately positioned the patient and is ready to start the scan.

What is the most important initial step the technologist has to apply?

- A. Start to acquire axial slice images
- B. Start to acquire AP scout image
- C. Start to acquire sagittal slice images
- D. Start to acquire coronal slice images

65. An abdominal CT exam was requested for a 45-year-old male patient with a history of alcoholism presenting with abdominal distension and breathing difficulty. The technologist was able to visualize, on the axial section, an organ that occupied the majority of the right upper quadrant showing variable appearance after intravenous contrast was injected.

What organ has the technologist observed?

- A. Head of the pancreas
- B. Inferior vena cava
- C. Gallbladder
- D. Liver

66. A 22-year-old soccer player was diagnosed with a posterior cruciate ligament tear. His physician ordered an MR exam of the knee joint to better visualize the tear.

What is be the most appropriate plane to demonstrate this tear?

- A. Axial plane
- B. Sagittal plane
- C. Oblique plane
- D. Coronal Plan

67. A 70-year-old male patient who is on chemotherapy for a diagnosed case of prostate cancer has come to the MR unit for examination of the Brain.

What is the most appropriate scanning protocol that has to be applied?

- A. Acquire pre-contrast DWI (Diffusion Weighted imaging) sequence only
- B. Acquire both pre and post contrast in all the major sequences
- C. Acquire post-contrast sequence in a single plane
- D. Acquire post-contrast T2 sequence only

68. A patient come for an MRI examination and through the screening process likely ferrous foreign bodies are reported to be near the orbits or vital organs. in order to make sure the existence of ferrous body what should the likely examination before the MRI

- A. CT/ PET
- B. X-ray/ CT Scan
- C. CT/ Ultrasound
- D. X-RAY/ Ultrasound

69. A 24 years old male Patient came for MRI exam after brain surgery for a control brain scan. He had his previous scan before his operation and it was done with contrast. The technologist in charge what would you do for the control

- A. Both T1 and T2 pre contrast sequences in all plane
- B. Both T1 and T2 pre and post contrast sequences in all plane
- C. Both T1 and T2 pre contrast, flair and diffusion sequences in all plane

- D. Both T1 and T2 pre and post contrast, flair and diffusion sequences in all plane
70. A 40 years old male known hypertensive patient came to the emergency department with loss of consciousness, left side body weakness, nausea and difficulty of speech. The physician ordered an MRI examination of the brain to r/o hemorrhagic stroke. Likely one have less contribution to r/o the hemorrhagic stroke
- A. T2-weighted imaging
 - B. T1 weighted contrast enhances
 - C. diffusion-weighted imaging (DWI)
 - D. Apparent diffusion coefficient (ADC)
71. A 40 years old male known hypertensive patient came to the emergency department with loss of consciousness, left side body weakness, nausea and difficulty of speech. The physician ordered an MRI examination of the brain to r/o hemorrhagic stroke. Which MRI finding confirms acute hemorrhagic stroke?
- A. T2 isointenseity
 - B. T1 hyperintense
 - C. T2 hyperintense
 - D. T2 hypointense with hyperintense perilesional rim
72. A patient comes for brain MRI contrast enhanced imaging During the contrast enhanced imaging technique his/he images r are produced where vessels signal intensity is primarily dependent on
- A. Blood composition
 - B. Cerebral blood volume
 - C. Patient body temperature
 - D. The velocity of flowing blood
73. A patient from orthopedic ward is sent to radiology department for a MRI examination, the patient has an orthopedic implant in his/her knee. What pulse sequences should be avoided in order to minimize magnetic susceptibility artifact
- A. Spin echo pulse sequences
 - B. Fast spin echo plus sequences

- C. Gradient echo plus sequences
- D. All pulse sequences produced the same amount of magnetic susceptibility artifact

74. A researcher wants to do countrywide research. Before going to start the data collection he did pilot study.

What is the purpose of doing this initial study?

- A. Draw sample
- B. Test the tools
- C. Identify problem
- D. Create hypothesis

75. MRT department wants to conduct a Research to find solution for the current immediate problem in the department

What is the most likely type of research for the above scenario?

- A. Fundamental research
- B. Analytical research
- C. Survey research
- D. Action research

76. A 45 years old male patient came to Arbaminch general Hospital with complain of back pain, claudication and impotency. The physician suspected abdominal aortic aneurism and sent him to radiology department for abdominal ultrasound. As medical radiologic technologist which region of abdominal aorta is commonest for aneurism and need special concentration during your scan?

- E. Infra renal
- F. Suprarenal
- G. At the Celiac axis
- H. At the common iliac artery

77. A 25 years old female patient came to radiology department with complain of acute onset of pelvic pain ultrasound showed well defined cystic mass with homogenous low level internal echo.

What is your diagnosis?

- A. Follicular cyst
- B. Hemorrhagic cyst
- C. Endometrium
- D. Endometritis

78. Which one of the following is anterior structure on ultrasound image of pancreatic head on transverse scan?

- A. Gastro colic trunk
- B. Common bile duct
- C. Superior mesenteric vein
- D. Superior mesenteric artery

79. A 40 years old male patient came to radiology department with complain of epigastric pain. What do you do when collapsed stomach mimics pancreatic duct on sonographic evaluation of the pancreas.?

- A. Letting your patient drink some water
- B. Appoint the patient for the next day with NPO
- C. Change the patient position from supine to prone
- D. None

80. A 35 years old male patient come to radiology department with history of epigastric pain nausea and vomiting. During your sonographic scanning what are you doing when you suspect fatty pancreas?

- A. Compare pancreatic parenchymal echogenicity with hepatic parenchymal echogenicity
- B. Compare pancreatic parenchymal echogenicity with spleen parenchymal echogenicity
- C. Recommend for CT scan
- D. Recommend for ERCP